Tufts University
SCHOOL OF ENGINEERING
Electrical and Computer Engineering

FPGA-based Laser Stabilization for Quantum Computing

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Introduction

Trapped-ion quantum computers use lasers to manipulate and stabilize atoms. These atoms encode information as qubits, which differ from bits by representing 0, 1, or a superposition of the two. These stabilizing lasers are highly sensitive to frequency drift and environmental noise. To address this, FPGAs/microcontrollers are used to stabilize and interface with the lasers. Current stabilization algorithms use high-level programming languages, introducing latency to the feedback loop. In collaboration with the University of Sydney Quantum Science Group and Professor Mark Hempstead, we aim to accelerate this process. By implementing the laser control logic in RTL (Register Transfer Level) code, we reduced feedback latency from 4ms to 1ms , mitigating long-term drift and improving short-term error correction.

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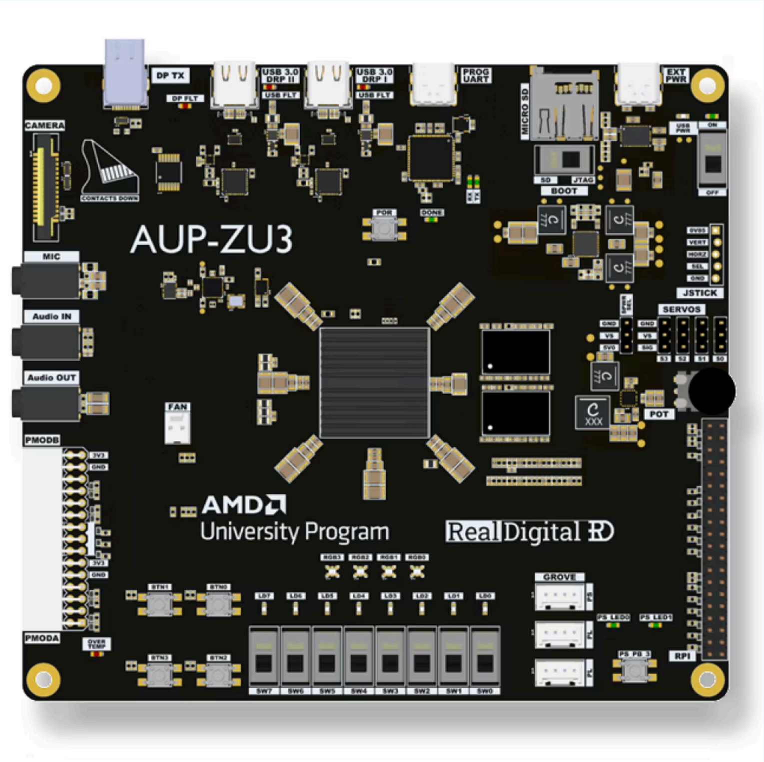
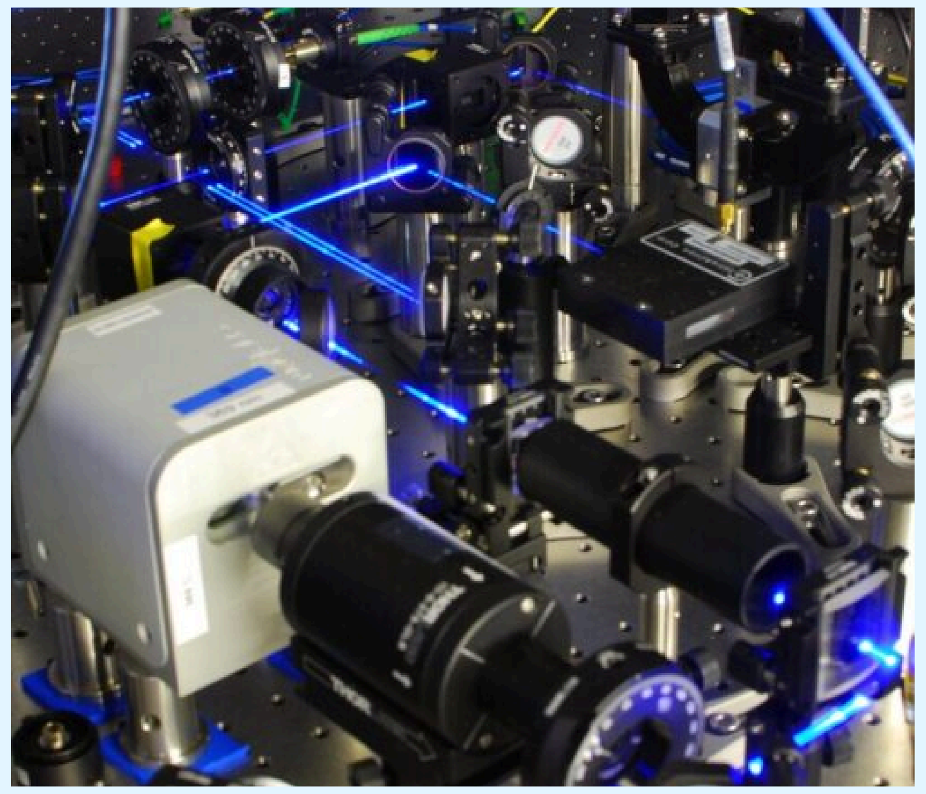




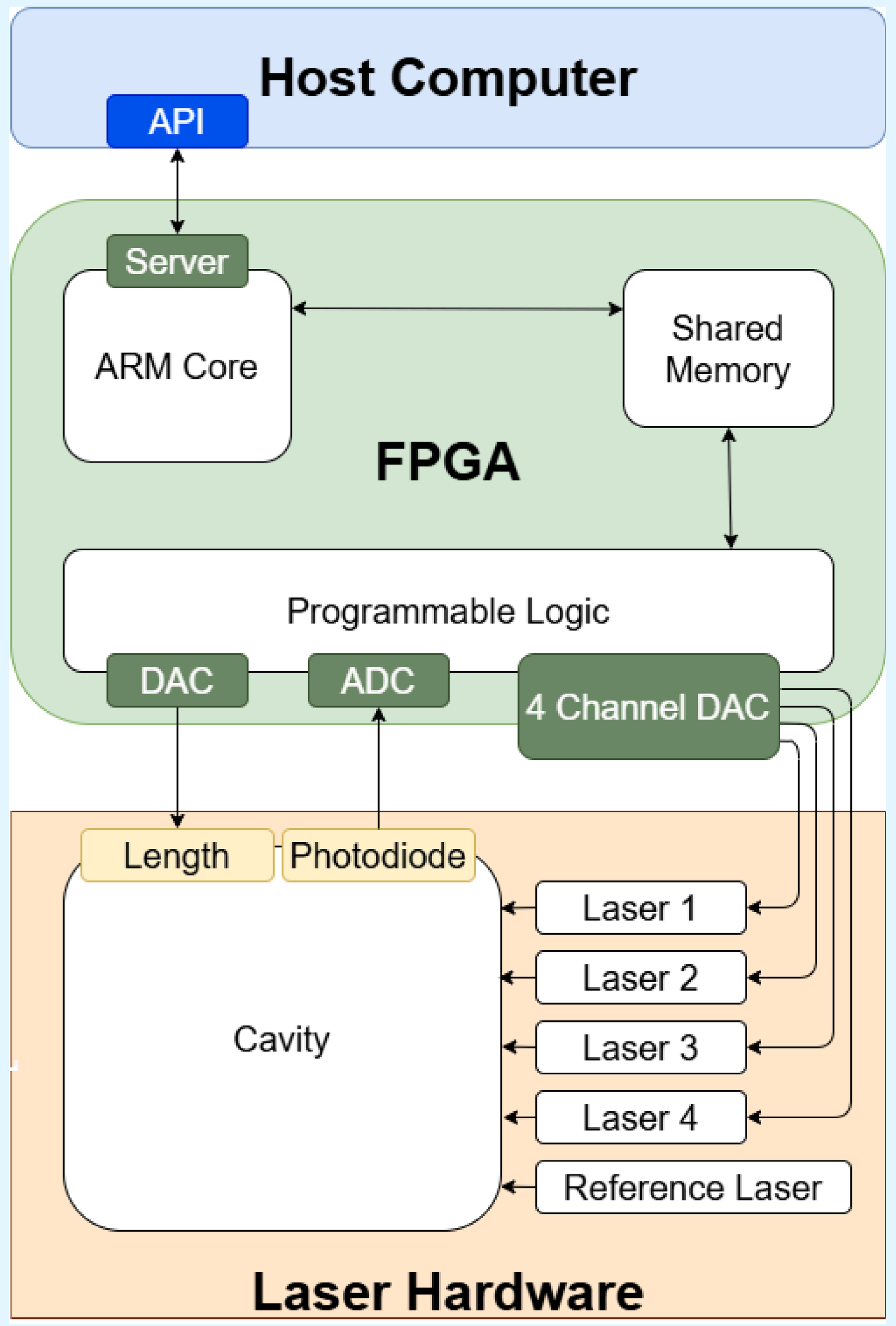
System Design

Our system requirements were to maintain the configurability and ease of use of existing solutions while building a faster, cheaper system.

To address the issue of speed, we decided to implement the real-time algorithms on an FPGA. The RealDigital AUP-ZU3 FPGA was selected due to its low cost and available I/O. We then selected the analog peripherals to be compatible with the FPGA and fast enough to improve over existing speeds. Due to our choice of FPGA and peripherals, we we able to keep all hardware on one board, a big improvement from existing solutions.

To maintain configurability and usability, we implemented a server on the FPGA and accompanying client API to enable simple, abstracted interaction with the system's modes and parameters.

Metrics	Our Solution	Existing Solution
Cost	$262	~$1800
Number of Lasers	4	4
Cavity Scan	1000 Hz	238 Hz
Laser Feedback	1000 Hz	100 Hz

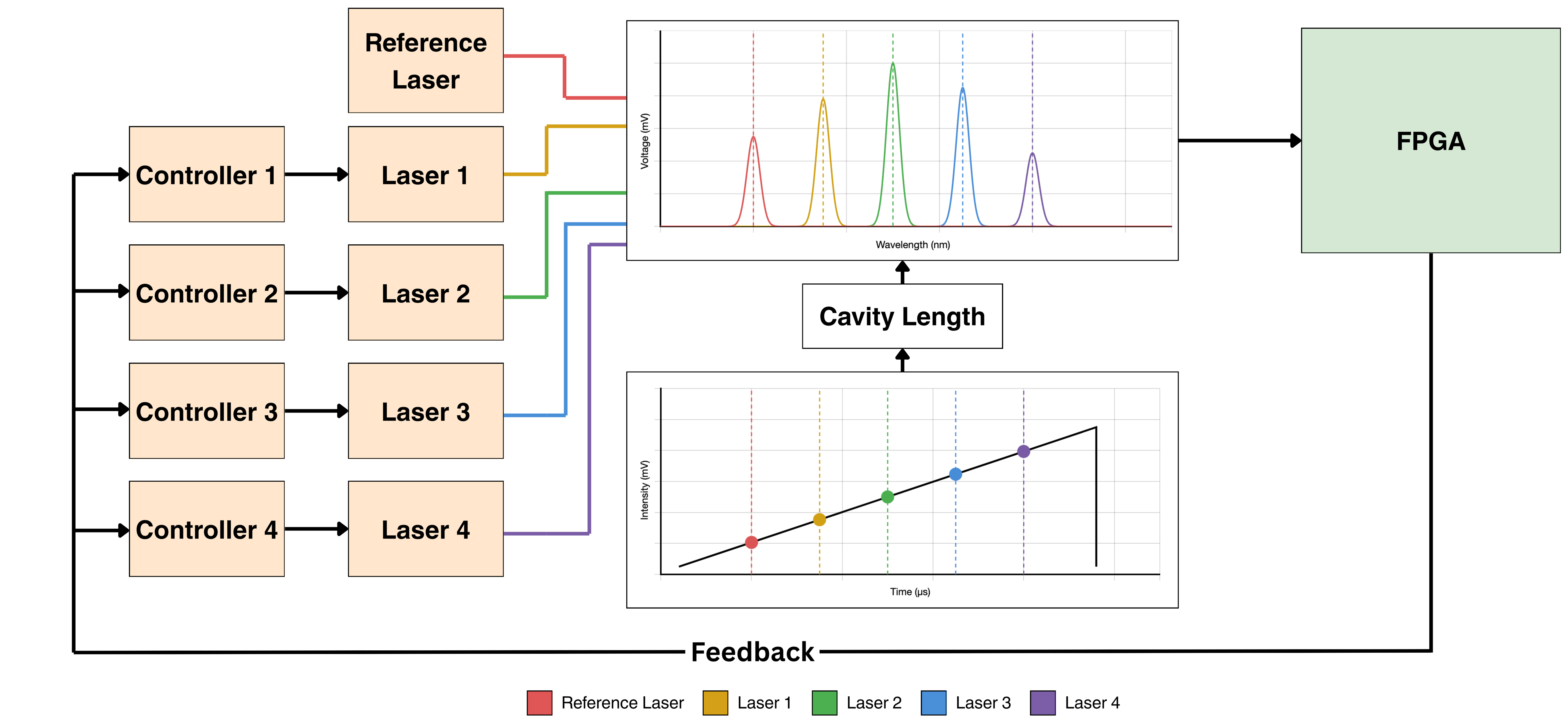


Laser Controllers
Each laser has a controller which takes in a voltage to adjust its wavelength.

Lasers
We support 4 lasers and 1 reference laser.
All 5 lasers are fed into the cavity.

Cavity
We sweep the length of the cavity to find the wavelengths of each laser, shown by spikes in intensity.

Algorithm
We figure out which peak belongs to which laser, then use a PID controller to adjust the feedback we provide to each laser controller.



Methods

Our system has the key advantage of implementing all of the peak detection and feedback control in RTL rather than in a high level language, significantly reducing overhead to allow for a faster feedback rate. We used an ARM core to handle data transfer between the board using a simple ethernet connection, creating a highly configurable and easy to program method of communication.

To characterize the lasers' frequencies, we sweep the width of a cavity while reading a photodiode, so each laser will have a characteristic frequency peak corresponding to a specific width. Our peak detection algorithm works by marking candidate voltage peaks when they cross a threshold, then assigning peaks to lasers that have matching target frequencies. To stabilize the laser, we provide feedback using a PID controller to minimize the difference between the detected and target frequencies.

We tested our algorithm with a simulation that incorporates long-term drift and sudden disturbances that Sydney's physical system experiences, including samples of their data. We simulated environmental noise, sudden frequency jumps, losing laser readings for a short period, and overlapping laser peaks. We then feed this data into our system through function generators and analyze performance through oscilloscopes and data dumps to the host computer.

Results

Our team developed a solution that is faster, less expensive, and more scalable than existing literature. Existing solutions rely on at least two FPGAs to achieve the desired results; this solution uses just one. Additionally, the Python-based API allows for greater accessibility for a wider range of users and seamless integration into overall laser control setup.

Acknowledgments

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[2] E. Pultinevicius et al., "A scalable scanning transfer cavity laser stabilization scheme based on the Red Pitaya STEMlab platform," Review of Scientific Instruments, vol. 94, no. 10, p. 103004, Oct. 2023, doi: 10.1063/5.0169021.